SREEANNAPURNA RALLABHANDI

AI/ML Engineer | Generative AI, LLM Applications & AI Agents | ML Deployment

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PROFESSIONAL SUMMARY

AI/ML Engineer with 6+ years across Python engineering, applied machine learning, and AI applications. Builds, trains, evaluates, and deploys ML models, and develops production-oriented LLM applications and AI agents. Built a 0-to-1 agentic AI analytics application with LangGraph, retrieval-augmented generation (RAG), FastAPI, and locally hosted Mistral 7B. Experienced in TensorFlow, scikit-learn, Vertex AI, inference APIs, model evaluation, Docker, and AWS, with a focus on reliability, accuracy, and scalability.

TECHNICAL SKILLS

Machine Learning & Deep Learning: Python, scikit-learn, TensorFlow, Keras, PyTorch, supervised and unsupervised ML, Random Forest, Isolation Forest, PCA, GRU, feature engineering, model validation

Generative AI & LLM Applications: LLMs, AI agents, LangGraph, LangChain, RAG, Mistral 7B, prompt engineering, tool calling, response grounding, LLM evaluation, reliability testing

Production ML & MLOps: Model deployment, inference APIs, FastAPI, REST APIs, Vertex AI, Docker, Kubernetes, AWS, GCP, CI/CD, Git, observability

Data & Analytics: SQL, Pandas, NumPy, PySpark, BigQuery, Snowflake, Airflow, Kafka, time-series forecasting, anomaly detection, TF-IDF, error analysis

PROFESSIONAL EXPERIENCE

Invences Inc **Dallas, TX**

AI Engineer – Agentic AI Analytics *Feb 2026 – Present*

- Built a 0-to-1 enterprise AI analytics application with Python, LangGraph, FastAPI, RAG, and locally hosted Mistral 7B that converts natural-language questions into data-grounded analytical responses.
- Developed REST inference and orchestration interfaces that coordinate agent state, data retrieval, tool selection, and response validation across operational, customer, and event data.
- Designed typed analytical tools and controlled execution paths that limit model-generated actions to approved data sources, calculations, and business rules.
- Implemented role-based access controls, query-level audit trails, and source-grounding checks so stakeholders can trace answers to retrieved evidence and authorized tool outputs.
- Developed LLM evaluation scenarios covering retrieval relevance, groundedness, tool-call correctness, response quality, and edge cases with QA and LLMOps teams.
- Added structured execution traces and error handling for agent decisions and tool inputs/outputs, and integrated Vector.dev, VictoriaMetrics, and Grafana for observability.

University of North Texas **Denton, TX**

Graduate Research Assistant – Applied AI/ML *Sep 2023 – May 2025*

- Built Python and PySpark preprocessing and feature-engineering pipelines for high-frequency UAV sensor telemetry, creating model-ready spatial, temporal, and environmental signals.
- Trained and validated supervised and unsupervised models for anomaly and flight-deviation detection, achieving up to 96% validation accuracy in research experiments.
- Applied Vertex AI and TensorFlow to classification and anomaly-detection workflows, accelerating model iteration.
- Containerized ML workloads with Docker and deployed models on AWS, using Kubernetes and CI/CD for repeatable training, testing, and releases.
- Built data integration pipelines processing 2M+ records daily from 10+ sources (Airflow, BigQuery), cutting analysis turnaround from six hours to 15 minutes.
- Developed Kafka, MongoDB, Flask, and REST API components connecting streaming telemetry ingestion to validated feature inputs and model experimentation.

Forsys Inc **Hyderabad, India**

Data Analyst II *Mar 2022 – Aug 2023*

- Built Python and SQL data services and ingestion pipelines for production analytics and downstream ML applications, with validation rules for feature and reporting quality.
- Automated reconciliation and exception workflows, cutting a manual reconciliation process from four hours to five minutes.
- Refined CRM data-quality controls with operational stakeholders, contributing to 15% better CRM accuracy and 60% fewer data-quality tickets.
- Analyzed sales-funnel trends with Python and SQL and shared findings that contributed to an 8% increase in conversion.

Vidcentum Software Pvt. Ltd. **Hyderabad, India**

Data Analyst *Jan 2020 – Dec 2021*

- Developed Python, SQL, and PySpark pipelines consolidating IoT and application sources into governed datasets for analytics and machine-learning use.
- Built a unified Snowflake and Redshift data layer standardizing outputs from 15+ systems, cutting reporting latency from 48 hours to under four hours.
- Added automated data-quality checks at ingestion checkpoints, reducing on-call incidents by 55%.

EDUCATION

M.S., Advanced Data Analytics (Artificial Intelligence Concentration), University of North Texas — Dec 2025
M.S., Interdisciplinary Studies (Applied Analytics), University of North Texas — Dec 2024
B.E., Computer Science Engineering, ACE Engineering College, Hyderabad — May 2019

CERTIFICATIONS

- Anthropic: Building with the Claude API · Claude Code in Action · Introduction to Model Context Protocol · Claude with Amazon Bedrock (Jul 2026)
- Business Analytics and Optimization, IIT Delhi (Jun 2022)